

The Impact of Hurricane Ian on the Unemployment Rate: Evidence from Lee County, FL

Glen E. Martin

Abstract

Lee County, Florida was devastated by Hurricane Ian on September 28, 2022, totaling an estimated \$3.22 billion in property damage. Using Local Area Unemployment Statistics, county-level controls, and a synthetic Lee County counterfactual, this paper found a 1.7 percentage point increase in the unemployment rate a month after impact, and the unemployment rate remained elevated until September 2023. These results contribute evidence of the short-term impacts of hurricanes on local labor markets, contributing to both the labor economics literature and the natural disaster literature.

Keywords: Hurricane Ian, Unemployment Rate, Synthetic Control Method

JEL Codes: R23, J64, Q54

1 Introduction

Hurricane Ian was a devastating Category 5 hurricane that made landfall in Southwest Florida on September 28, 2022, causing catastrophic storm surge, flooding, and wind damage to 24,585 residential structures and 4,690 commercial structures, totaling an estimated \$3.22 billion in property loss in Lee County (Lee County Emergency Management, 2023). Hurricanes are documented to disrupt the local labor market by increasing the unemployment rate (Schwam and Coe, 2023; Scott et al., 2023) and decreasing employment (Belasen and Polachek, 2008, 2009). The current economic literature on Hurricane Ian has been related to its impact on housing prices in Lee County (Contat et al., 2024). At present, no study has investigated the impact of Hurricane Ian on the labor market in Lee County.

The synthetic control methodology is being used in a growing body of literature to examine the economic impact of hurricanes (Coffman and Noy, 2012; Schwam and Coe, 2023; Rayamajhee et al., 2024; Contat et al., 2024) and other natural disasters (Cavallo et al., 2013; Barone and Mocetti, 2014; Best and Burke, 2019). This paper examines the short-term impact of Hurricane Ian on Lee County’s monthly unemployment rate using the synthetic control method pioneered by Abadie et al. (2010, 2015) and finds that Hurricane Ian increased the unemployment rate in Lee County by approximately 1.7 percentage points in October 2022. The unemployment rate remained elevated for eleven months until it returned to pre-hurricane levels by September 2023.

The remainder of the paper is organized as follows. Section 2 delineates the data used in the analysis. Section 3 discusses the synthetic control methodology through the lens of the potential outcomes framework. Section 4 provides the synthetic control results for Lee County and the placebo analysis. Section 5 concludes by summarizing the results, addressing limitations, and providing future research direction.

2 Data

The synthetic control analysis used the monthly unemployment rate from the Bureau of Labor Statistics’ Local Area Unemployment Statistics (LAUS) as the outcome of interest. The analysis is limited to a January 2021 to December 2023 time frame to avoid potential confounding effects from Hurricane Irma (2017) and COVID-19 (2020).

The unemployment rate was lagged at the 1-month level and 1-year level to use as additional covariates in the model. In addition to lagged outcomes, other covariates included log RGDP per capita (Bureau of Economic Analysis), population growth rates (U.S. Census) and demographic data from the American Community Survey 5-year estimates. These covariates were lagged by 1-year since these sources are only available at a yearly level to avoid contamination from Hurricane Ian.

The donor pool was constructed using County Adjacency data from the National Bureau of Economic Research to determine county neighbors, and the Storm Events Database to determine hurricane exposure.

3 Synthetic Control Method

Let UNR_{jt} denote the unemployment rate, the outcome interest, for unit j of $J+1$ at month t where the unit of interest, Lee County, is $j = 1$, and $j = 2, \dots, J+1$ are potential units of comparison. Suppose D is a 1/0 indicator on whether a county is impacted by Hurricane Ian. Based on the causal analysis potential outcomes framework described in Cunningham (2021), the unemployment rate of Lee County is a function of the switching equation of the potential outcomes.

$$UNR_{1t} = UNR_{1t}^1 D_i + UNR_{1t}^0 (1 - D_i)$$

where UNR_{1t}^1 is the unemployment rate if the hurricane impacts Lee County and UNR_{1t}^0

is the outcome if the hurricane did not impact Lee county. We are interested in knowing the unit-specific treatment effect of Hurricane Ian for Lee County.

$$\tau_{1,t} = UNR_{1t}^1 - UNR_{1t}^0$$

UNR_{1t}^0 is unobservable, however, we can estimate it through the synthetic control method by estimating the unemployment rate as a weighted average of optimally chosen units that the hurricane did not impact as the synthetic control.

$$\widehat{UNR_{1t}^0} = \sum_{j=2}^{J+1} w_j^* UNR_{jt}^0$$

where w is a vector of weights chosen by an algorithm (Abadie et al., 2010, 2015; Abadie, 2021). The synthetic control estimator measures the unit-specific treatment effect in the post-intervention period as

$$\hat{\tau}_{1,t} = UNR_{1t}^1 - \widehat{UNR_{1t}^0}$$

The matching variables X_1 and X_0 are chosen predictors of the outcome and cannot be affected by the intervention. The weights are chosen to minimize the norm $\|X_1 X_0 W\|$ subject to the weight constraints. Let W be a $(J \times 1)$ vector of positive weights that sum to one, $W = (w_2, \dots, w_{J+1})'$ with $w_j \geq 0$ and $w_2 + \dots + w_{J+1} = 1$.

The synthetic control weights minimize:

$$\sum_{m=1}^k V_m (X_{1m} - \sum_{j=2}^{J+1} W_j X_{jm})^2$$

where V_m is a weight that reflects the relative importance assigned to the m^{th} variable when measuring the difference between the treated unit and the synthetic control. V is important because W depends on the choice of V . The synthetic control $W(V)$ simulates the behavior of the outcome variable for the treated unit if the intervention never occurred.

The weights $v_1, \dots, v_k$ should reflect the predicted weights of the covariates. Abadie et al. (2010, 2015) suggests that different choices of V can be used, however, Cunningham (2021) points out in practice V is chosen that minimizes the pre-treatment mean squared prediction error (RMSPE).

3.1 Choosing the Donor Pool

As Abadie (2021) pointed out if the donors in the model J are larger than the number of pre-intervention time periods T_0 we run the risk of overfitting the model and biasing the results. This is a particular concern on the county level because in Florida there are 67 counties while $T_0 = 20$ ¹. The donor pool J is constructed from the universe of donors U_D by using the following criteria to help mitigate these concerns.

Units in U_D are not included in J if they were impacted by Hurricane Ian in 2022, adjacent to the unit of interest, ie., Lee County, and are not closely related to the unit of interest from a K-Means cluster analysis

We used the mean of the covariates and unemployment rate in periods before the intervention in the K-Means cluster analysis. A value of K was chosen based on the elbow method while also trying to limit the size of the donor pool so it is not much larger than the number of time periods.

¹Only Florida counties in the universe of donors to ensure there are no state level differences that need to be controlled for (Schwam and Coe, 2023).

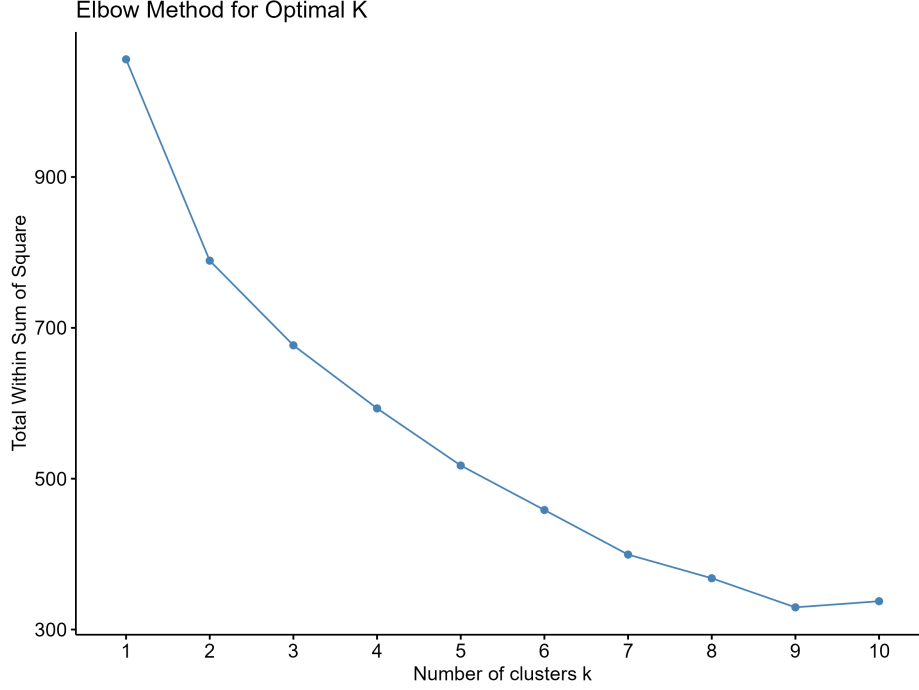


Figure 1: Elbow Plot for Optimal K

Figure 1 suggests that the number of clusters should be $K = 2$, however this would give us a donor pool of 31 counties, which may cause the model to overfit the data. Therefore, K was increased until the number of counties were close to T_0 . Both $K = 3$ and $K = 4$ gave a list of donors that was closer to the fixed time period. $K = 3$ was used since it yielded donor pool of 22 counties and is closer to the fixed time period than $K = 4$ which yielded a donor pool of 11 counties. Table 1 (pg. 11), shows the composition of counties for each value of K .

Drawing on prior research, this paper performed an in-space and in-time placebo analysis (Abadie et al., 2010, 2015; Abadie, 2021; Gilchrist et al., 2023).

In-space placebo analysis involves iteratively reassigning the treatment county to one of the counties in the donor pool and including the actual treated unit in the false treated unit's donor pool. After running the models for each county in the donor pool, we can calculate the ratio of post-period to pre-period root mean squared predictive error (RMSPE) for each county, which is used to generate a p-value. The results are robust if the ratio of the post-period to pre-period RMSPEs in Lee County is the most extreme compared to the other

counties.

The in-time placebo analysis is a falsification test to see if the effect of the intervention is robust to a false start. This method adjusts the intervention date to before or after the actual intervention date in the synthetic control analysis with the same treatment unit. The results are robust when there is not a divergence between the results of the initial analysis and the in-time placebo analysis. A requirement of the in-time placebo analysis is to have a sufficiently large number of periods where no other structural shocks to the outcome occur (Abadie et al., 2015). In contrast to the in-space placebo analysis, there are no p-values to evaluate the in-time placebo analysis.

4 Results

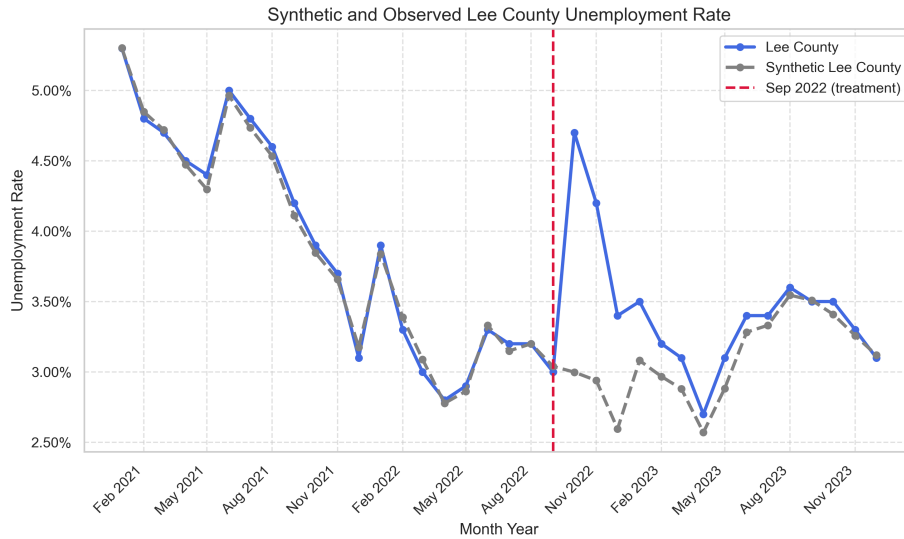


Figure 2: Synthetic Control Plot

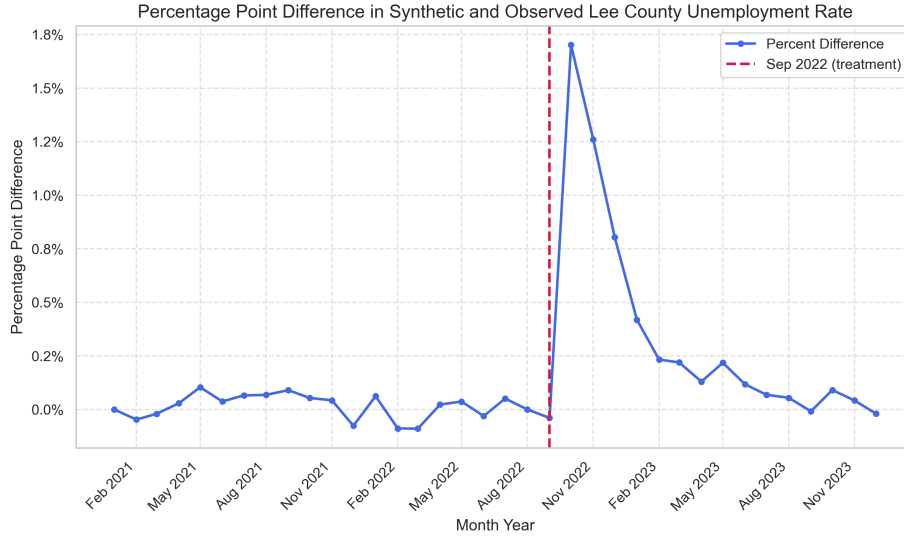


Figure 3: Synthetic Control Difference Plot

Figure 2 presents the impact of Hurricane Ian on the unemployment rate in Lee County. The evidence indicates a six-month-long spike in the unemployment rate that declined over time to the counterfactual unemployment rate. As Figure 3 indicates, the unemployment rate due to Hurricane Ian was about 1.7 percentage points higher than the counterfactual unemployment rate in October 2022. Although the largest impact to the unemployment rate occurred in the first six months after the hurricane’s landing, the unemployment rate in Lee County remained higher than the counterfactual for eleven months and did not reach pre-Hurricane Ian levels until September 2023.

Although geographically diverse, they contributed to Lee County’s synthetic control due to their similar demographic and economic characteristics. Lee County is best synthesized as a linear combination of St. Johns (27%), Monroe (23%), Polk (22%), Citrus(13%), Sumter (7%), St. Lucie (5%), and Hernando (2%), respectively.

Figure 4 shows a bar chart of the ratio of post/pre RMSPEs for Lee County and each county in the donor pool. Lee County ranked number 1 which produces a p-value less than 5% ($1/23=0.04$). Therefore, we reject the null hypothesis of no effect in favor of the alternative that Hurricane Ian had an effect on the unemployment rate in Lee County.

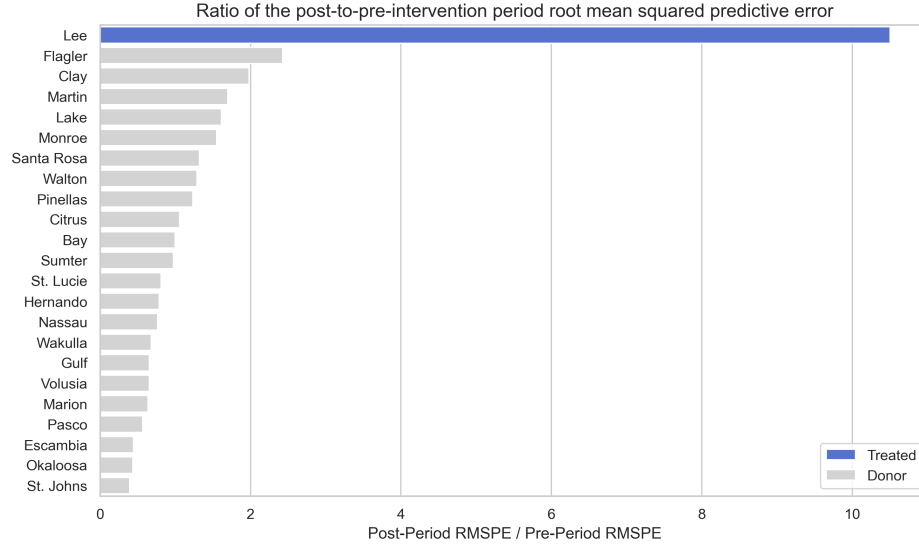


Figure 4: In-Space Placebo Significance

Following Gilchrist et al. (2023) the Zivot-Andrews test is used to provide a data-driven alternative for the intervention date in the in-time placebo analysis. We subset the time-series of the unemployment rate in Lee County to exclude the months after the September 2022 intervention. The estimate of the Zivot-Andrews test pointed to August 2022 as a period where a structural break in the time series is perceptible, so we used this as the false start date in the in-time placebo analysis.

Figure 5 reports the results of the in-time placebo analysis. The in-time placebo analysis almost perfectly replicates the findings of Figure 2 suggesting there is no evidence of a structural break comparable to Hurricane Ian in September 2022. The initial results are robust to the in-time placebo analysis because there is no divergence at the false date.

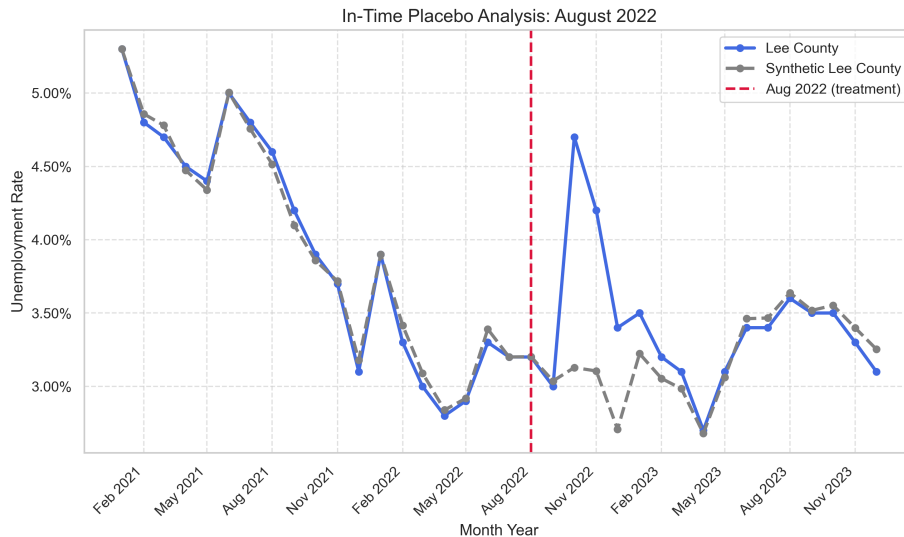


Figure 5: In-Time Placebo Plot

5 Conclusion

Although there is a growing body of literature that analyzes the economic impacts of hurricanes, little research has been done to assess the impact that Hurricane Ian had on local labor markets. In this paper, Lee County is used as a case study to estimate the Hurricane Ian's impact on the local unemployment rate. The results of the synthetic control analysis indicate that Hurricane Ian increased the unemployment rate in the following month after the hurricane landed and remained elevated for eleven additional months. These results are congruent with research for the labor market impact of Hurricane Micheal which found that affected counties had higher unemployment rates for six to eleven months after the hurricane (Scott et al., 2023).

As Yun and Kim (2022) pointed out, spatial aggregation bias may be present if the improper spatial scale is used. The results at the county level may potentially underestimate the impact Hurricane Ian had on the local unemployment rate. Future research can address this by repeating the analysis at the metropolitan statistical area (MSA) level and zip code level. Finding data at the zip code level nevertheless is challenging for a monthly frequency due to confidentiality restrictions. While a lower spatial scale may provide a more accurate estimate of the local area treatment effect, the time scale used is also important. Time aggregations at the quarterly or annual level may not capture the impact hurricanes have on labor markets if the impact is short-term.

Future research on the impact of Hurricane Ian on local labor markets should address the following questions: "What was the overall impact of Hurricane Ian on the unemployment rate?" and "What was the impact of Hurricane Ian on employment across industries?" The former question addresses the overall effect Hurricane Ian had on the county level, while the latter question addresses the heterogeneous impact hurricanes can have across different industries.

References

- Abadie, A. (2021). Using Synthetic Controls: Feasibility, Data Requirements, and Methodological Aspects. *Journal of Economic Literature*, 59(2):391–425.
- Abadie, A., Diamond, A., and Hainmueller, J. (2010). Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California’s Tobacco Control Program. *Journal of the American Statistical Association*, 105(490):493–505.
- Abadie, A., Diamond, A., and Hainmueller, J. (2015). Comparative Politics and the Synthetic Control Method. *American Journal of Political Science*, 59(2):495–510.
- Barone, G. and Mocetti, S. (2014). Natural Disasters, Growth and Institutions: a Tale of Two Earthquakes. *Journal of Urban Economics*, 84:52–66.
- Belasen, A. R. and Polachek, S. W. (2008). How Hurricanes Affect Wages and Employment in Local Labor Markets. *American Economic Review*, 98(2):49–53.
- Belasen, A. R. and Polachek, S. W. (2009). How Disasters Affect Local Labor Markets: The Effects of Hurricanes in Florida. *Journal of Human Resources*, 44(1):251–276.
- Best, R. and Burke, P. J. (2019). Macroeconomic Impacts of the 2010 Earthquake in Haiti. *Empirical Economics*, 56(5):1647–1681.
- Cavallo, E., Galiani, S., Noy, H., and Pantano, J. (2013). Catastrophic Natural Disasters and Economic Growth. *The Review of Economics and Statistics*, 95(5):1549–1561.
- Coffman, M. and Noy, I. (2012). Hurricane Iniki: Measuring the Long-Term Economic Impact of a Natural Disaster Using Synthetic Control. *Environment and Development Economics*, 17(2):187–205.
- Contat, J. C., Doerner, W. M., Renner, R. N., and Rogers, M. J. (2024). Measuring Price Effects from Disasters Using Public Data: A Case Study of Hurricane Ian. *Journal of Real Estate Research*, pages 1–48.
- Cunningham, S. (2021). *Causal Inference: The Mixtape*. Yale University Press, New Haven, Connecticut.
- Gilchrist, D., Emery, T., Garoupa, N., and Spruk, R. (2023). Synthetic Control Method: A Tool for Comparative Case Studies in Economic History. *Journal of Economic Surveys*, 37(2):409–445.
- Lee County Emergency Managment (2023). Hurricane Ian After-Action Report Lee County.
- Rayamajhee, V., March, R. J., and Clark, C. C. T. (2024). Shock me like a Hurricane: how Hurricane Katrina changed Louisiana’s formal and Informal Institutions. *Journal of Institutional Economics*, 20:1–20.
- Schwam, D. and Coe, J. (2023). Estimating the Local Labor Market Effects of Hurricanes. *SSRN Electronic Journal*.
- Scott, B. L., Thomas, N., Kirby, R. S., Reader, S., L Merlo, K., and Marshall, J. (2023). Employment Impacts and Industry Workforce Shifts in the Florida Panhandle Post-Hurricane Michael. *International Journal of Mass Emergencies & Disasters*, 41(1):85–93.
- Yun, S. D. and Kim, A. (2022). Economic Impact of Natural Disasters: a Myth or Mismeasurement? *Applied Economics Letters*, 29(10):861–866.

Tables

Table 1: Donor Pools from Clusters

K=2	K=3	K=4
Alachua	Bay	
Bay		
Broward		
Citrus	Citrus	Citrus
Clay	Clay	
Duval		
Escambia	Escambia	
Flagler	Flagler	Flagler
	Gulf	
Hernando	Hernando	Hernando
Highlands		Highlands
Hillsborough		
Lake	Lake	Lake
Leon		
Marion	Marion	Marion
Martin	Martin	
Miami-Dade		
Monroe	Monroe	
Nassau	Nassau	
Okaloosa	Okaloosa	
Orange		
Palm Beach		
Pasco	Pasco	Pasco
Pinellas	Pinellas	Polk
Polk		
Santa Rosa	Santa Rosa	
Seminole	St. Johns	
St. Johns	St. Lucie	St. Lucie
St. Lucie	Sumter	Sumter
Sumter	Volusia	Volusia
Volusia	Wakulla	
Walton	Walton	